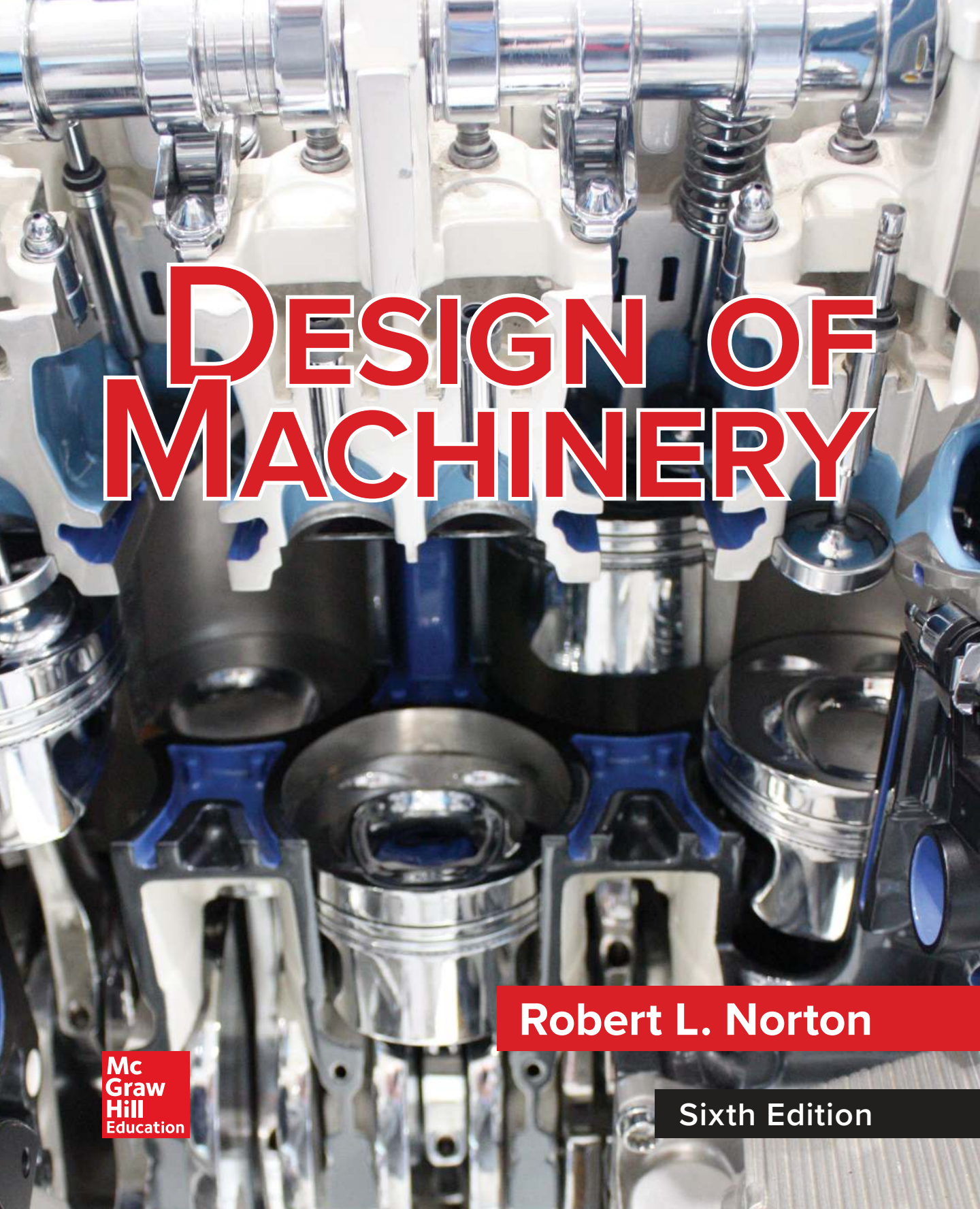




DESIGN OF MACHINERY

Robert L. Norton

**Mc
Graw
Hill**
Education

Sixth Edition

DESIGN OF MACHINERY

AN INTRODUCTION TO THE SYNTHESIS AND
ANALYSIS OF MECHANISMS
AND MACHINES

Sixth Edition



McGraw-Hill Series in Mechanical Engineering

Alciatore	<i>Introduction to Mechatronics and Measurement Systems</i>
Anderson	<i>Fundamentals of Aerodynamics</i>
Anderson	<i>Introduction to Flight</i>
Anderson	<i>Modern Compressible Flow</i>
Beer/Johnston	<i>Vector Mechanics for Engineers</i>
Beer/Johnston	<i>Mechanics of Materials</i>
Budynas	<i>Advanced Strength and Applied Stress Analysis</i>
Budynas/Nisbett	<i>Shigley's Mechanical Engineering Design</i>
Byers/Dorf/Nelson	<i>Technology Ventures: From Idea to Enterprise</i>
Cengel	<i>Introduction to Thermodynamics and Heat Transfer</i>
Cengel/Boles	<i>Thermodynamics: An Engineering Approach</i>
Cengel/Cimbala	<i>Fluid Mechanics: Fundamentals and Applications</i>
Cengel/Ghajar	<i>Heat and Mass Transfer: Fundamentals and Applications</i>
Cengel/Turner/Cimbala	<i>Fundamentals of Thermal-Fluid Sciences</i>
Dieter/Schmidt	<i>Engineering Design</i>
Finnemore/Franzini	<i>Fluid Mechanics with Engineering Applications</i>
Heywood	<i>Internal Combustion Engine Fundamentals</i>
Holman	<i>Experimental Methods for Engineers</i>
Holman	<i>Heat Transfer</i>
Kays/Crawford/Weigand	<i>Convective Heat and Mass Transfer</i>
Norton	<i>Design of Machinery</i>
Palm	<i>System Dynamics</i>
Plesha/Gray/Costanzo	<i>Engineering Mechanics: Statics and Dynamics</i>
Reddy	<i>An Introduction to the Finite Element Method</i>
Schey	<i>Introduction to Manufacturing Processes</i>
Smith/Hashemi	<i>Foundations of Materials Science and Engineering</i>
Turns	<i>An Introduction to Combustion: Concepts and Applications</i>
Ullman	<i>The Mechanical Design Process</i>
White	<i>Fluid Mechanics</i>
White	<i>Viscous Fluid Flow</i>
Zeid	<i>Mastering CAD/CAM</i>

DESIGN OF MACHINERY

AN INTRODUCTION TO THE SYNTHESIS AND ANALYSIS OF MECHANISMS AND MACHINES

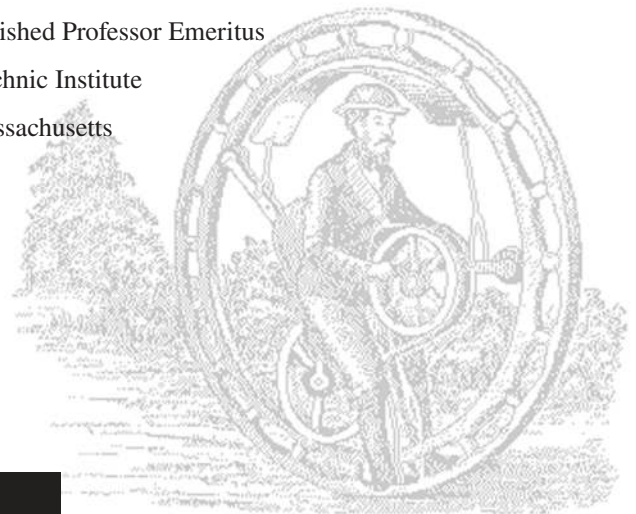
Sixth Edition

Robert L. Norton

Milton P. Higgins II Distinguished Professor Emeritus

Worcester Polytechnic Institute

Worcester, Massachusetts



**Mc
Graw
Hill
Education**



DESIGN OF MACHINERY: An Introduction to the Synthesis and Analysis of Mechanisms and Machines, Sixth Edition

Published by McGraw-Hill Education, 2 Penn Plaza, New York, NY, 10121. Copyright © 2020 by McGraw-Hill Education. All rights reserved. Printed in the United States of America. Previous editions © 2012, 2008, 2004, 2001, 1999, and 1992. No part of this publication may be reproduced or distributed in any form or by any means, or stored in a database or retrieval system, without the prior written consent of McGraw-Hill Education, including, but not limited to, in any network or other electronic storage or transmission, or broadcast for distance learning.

Some ancillaries, including electronic and print components, may not be available to customers outside the United States.

This book is printed on acid-free paper containing 10% postconsumer waste.

1 2 3 4 5 6 7 8 9 LCR 21 20 19

ISBN 978-1-260-11331-0 (bound edition)

MHID 1-260-11331-0 (bound edition)

ISBN 978-1-260-43130-8X (loose-leaf edition)

MHID 1-260-43130-4 (loose-leaf edition)

Portfolio Manager: Thomas Scaife, Ph.D.

Product Developers: Tina Bower; Megan Platt

Marketing Manager: Shannon O'Donnell

Content Project Managers: Jeni McAttee; Rachael Hillebrand

Buyer: Susan K. Culbertson

Design: Debra Kubiak

Content Licensing Specialist: Melissa Homer

Cover Image and Design: ©Robert L. Norton

Compositor: Robert L. Norton and Aptara, Inc.

All credits appearing on page or at the end of the book are considered to be an extension of the copyright page.

All text, drawings, and equations in this print book were prepared and typeset electronically, by the author, on a *Macintosh*® computer using *Illustrator*®, *MathType*®, and *InDesign*® desktop publishing software. The body text is set in Stix General, and headings set in Proxima Nova. Color-separated printing plates were made on a direct-to-plate laser typesetter from the author's files. All *clip art* illustrations are courtesy of *Dubl-Click Software Inc.*, 22521 Styles St., Woodland Hills CA 91367, reprinted from their *Industrial Revolution* and *Old Earth Almanac* series with their permission (and with the author's thanks). The cover photo, by the author, is a cut-away of a VR-15 six-cylinder engine.

Library of Congress Cataloging-in-Publication Data

Names: Norton, Robert L., Author

Title: Design of machinery : an introduction to the synthesis and analysis of mechanisms and machines / Robert L. Norton.

Description: Sixth edition. | New York, NY : McGraw-Hill, [2020] | Includes bibliographical references and index.

Identifiers: LCCN 2018034424 | ISBN 9781260113310 (alk. paper) | ISBN 1260113310 (alk. paper)

Subjects: LCSH: Machine design. | Machinery, Kinematics of. | Machinery, Dynamics of.

Classification: LCC TJ230 .N63 2020 | DDC 621.8/15--dc23 LC record available at <https://lccn.loc.gov/2018034424>

The Internet addresses listed in the text were accurate at the time of publication. The inclusion of a website does not indicate an endorsement by the authors or McGraw-Hill Education, and McGraw-Hill Education does not guarantee the accuracy of the information presented at these sites.

Publisher's web page is mheducation.com/highered. Author's web page is <http://www.designofmachinery.com>

ABOUT THE AUTHOR

Robert L. Norton earned undergraduate degrees in both mechanical engineering and industrial technology at Northeastern University and an MS in engineering design at Tufts University. He was awarded a Dr. Eng. (h.c.) by WPI in 2012. He is a registered professional engineer in Massachusetts and Florida. He has extensive industrial experience in engineering design and manufacturing and many years' experience teaching mechanical engineering, engineering design, computer science, and related subjects at Northeastern University, Tufts University, and Worcester Polytechnic Institute.

At Polaroid Corporation for 10 years, he designed cameras, related mechanisms, and high-speed automated machinery. He spent three years at Jet Spray Cooler Inc., designing food-handling machinery and products. For five years he helped develop artificial-heart and noninvasive assisted-circulation (counterpulsation) devices at the Tufts New England Medical Center and Boston City Hospital. Since leaving industry to join academia, he has continued as an independent consultant on engineering projects ranging from disposable medical products to high-speed production machinery. He holds 13 U.S. patents.

Norton has been on the faculty of Worcester Polytechnic Institute since 1981 and is currently Milton Prince Higgins II Distinguished Professor Emeritus in the mechanical engineering department. He taught undergraduate and graduate courses in mechanical engineering with an emphasis on design, kinematics, vibrations, and dynamics of machinery for 31 years at WPI before retiring from active teaching.

He is the author of numerous technical papers and journal articles covering kinematics, dynamics of machinery, cam design and manufacturing, computers in education, and engineering education and of the texts *Machine Design: An Integrated Approach*, 6ed and the *Cam Design and Manufacturing Handbook*, 2ed. He is a Fellow and life member of the American Society of Mechanical Engineers and a past member of the Society of Automotive Engineers. In 2007, he was chosen as *U. S. Professor of the Year* for the State of Massachusetts by the *Council for the Advancement and Support of Education (CASE)* and the *Carnegie Foundation for the Advancement of Teaching*, who jointly present the only national awards for teaching excellence given in the United States of America.

WHAT USERS SAY ABOUT THE BOOK

Your text is the best of all the texts I have used—the balance of fundamentals and practice is especially important, and you have achieved that with aplomb!

—Professor John P. H. Steele, *Colorado School of Mines*

We picked your book (years ago) because it was (and still is) the most accessible undergraduate presentation of the material I have found; although clearly theoretically based (and not afraid of the math!), you have a style and method that brings the material alive to students.

—Professor Michael Keefe, *University of Delaware*

*As an instructor who has been using *Design of Machinery* in my classes for over 12 years, I've been especially impressed throughout by Professor Norton's care and attention in support of its users.*

—Professor John Lee, *San Diego State University*

This book is dedicated to the memory of my father,

Harry J. Norton, Sr.

who sparked a young boy's interest in engineering;

to the memory of my mother,

Kathryn T. Norton

who made it all possible;

to my wife,

Nancy Norton

who provides unflagging patience and support;

and to my children,

Robert, Mary, and Thomas,

who make it all worthwhile.

PREFACE

to the Sixth Edition

The sixth edition is an evolutionary improvement over the fifth and earlier editions. See the updated *Preface to the First Edition* (overleaf) for more detailed information on the book's purpose and organization. The principal changes in this edition are:

- In addition to the printed version of the text, digital e-book versions are also available. These have hotlinks to all the videos and to the downloadable content provided. There are 188 videos. All of these are marked in the print version as well, with their URLs provided, and they can be downloaded by print-book users. A **Video Contents** is provided, and all other downloadable items are listed in the **Downloads Index**.
- Over 50 new problem assignments have been added. The problem figures are included as downloadable PDF files so that students can easily print hard copies on which to work the solutions.
- The author-written programs that come with the book have been completely rewritten to improve their interface and usability, and they are now compatible with the latest operating systems and computers. The programs FOURBAR, FIVEBAR, SIXBAR, SLIDER, and ENGINE have been combined in a new program called LINKAGES that does everything those programs collectively did with new features added. Program DYNACAM also has been completely rewritten and is much improved. Program MATRIX is updated. These computer programs undergo frequent revision to add features and enhancements. Professors who adopt the book for a course and students using the print book may register to download the latest student versions of these programs from: <http://www.designofmachinery.com>. Click on the *Student* or *Professor* link.
- The *Working Model* program is needed to run the Working Model files included with this text. Some universities have site licenses for this program on their lab computers. The supplier, *Design Simulation Technologies*, offers student licenses for one-semester or one-year periods at moderate cost. These are available at <http://www.design-simulation.com/Purchase/studentproducts.php>.
- Many small improvements have been made to the discussion of a variety of topics in many chapters, based largely on user feedback, and all known errors have been corrected.

The extensive DVD content that was introduced in the Fifth Edition is now downloadable from a website. These downloads include:

- The entire *Hrones and Nelson Atlas of Coupler Curves* and the *Zhang et al Atlas of Geared Fivebar Coupler Curves*.
- Wang's *Mechanism Simulation in a Multimedia Environment* contains 105 Working Model (WM) files based on the book's figures with AVI files and 19 Matlab® models for kinematic analysis and animation. The AVI files are linked to their figures in the e-books.
- Videos of two "virtual laboratories" that replicate labs created by the author at WPI are provided. These include demonstrations of the lab machines used and spreadsheet files of the acceleration and force data taken during the experiments. The intent is to allow students at other schools to do these exercises as virtual laboratories.

- A series of 34 *Master Lecture Videos* by the author that cover most of the topics in the book as well as 39 shorter "snippets" from these lectures are woven into the chapters. Seven *Demonstration Videos* are also provided. These were recorded over the author's thirty-one years of teaching these subjects at WPI and are listed in the **Video Contents**.

All the downloadable files are accessible to digital-book users through the publisher's website via links in the digital book. Any instructor or student who uses the print book may register on my website, <http://www.designofmachinery.com>, either as a student or instructor, and I will send them a password to access a protected site where they can download the latest versions of my computer programs, LINKAGES, DYNACAM, and MATRIX, all videos, and all files listed in the Downloads Index. Note that I personally review each of these requests for access and approve only those that are filled out completely and correctly according to the provided instructions. I require complete information and only accept university email addresses.

ACKNOWLEDGMENTS The sources of photographs and other nonoriginal art used in the text are acknowledged in the captions and opposite the title page, but the author would also like to express his thanks for the cooperation of all those individuals and companies who generously made these items available. The author is indebted to, and would like to thank, a number of users who kindly notified him of errors or suggested improvements in all editions since the first. These include: Professors *Chad O'Neal* of Louisiana Tech, *Bram Demeulenaere* of Leuven University, *Eben Cobb* of WPI, *Diego Galuzzi* of University of Buenos Aires, *John R. Hall* of WPI, *Shafik Iskander* of U. Tennessee, *Richard Jakubek* of RPI, *Cheong Gill-Jeong* of Wonkwang University, Korea, *Swami Karunamoorthy* of St. Louis University, *Pierre Larochelle* of Florida Tech, *Scott Openshaw* of Iowa State, *Francis H. Raven* of Notre Dame, *Arnold E. Sikkema* of Dordt College, and *Donald A. Smith* of U. Wyoming.

Professors *M. R. Corley* of Louisiana Tech, *R. Devashier* of U. Evansville, *K. Gupta* of U. Illinois-Chicago, *M. Keefe* of U. Delaware, *J. Steffen* of Valparaiso University, *D. Walcerz* of York College, and *L. Wells* of U. Texas at Tyler provided useful suggestions or corrections. Professors *L. L. Howell* of BYU, and *G. K. Ananthasuresh* of U. Penn supplied photographs of compliant mechanisms. Professor *C. Furlong* of WPI generously supplied MEMS photos and information. Special thanks to *James Cormier* and *David Taranto* of WPI's Academic Technology Center for their help in creating the videos.

The author would like to express his appreciation to Professor *Sid Wang* of NCAT and his students for their efforts in creating the *Working Model* and *Matlab* files. Professor *Thomas A. Cook*, Mercer University (Emeritus), provided most of the new problem sets as well as their solutions in his impressive and voluminous solutions manual and its accompanying *Mathcad*® solution files. The author is most grateful for Dr. Cook's valuable contributions. Most of all, the author again thanks his infinitely patient wife, Nancy, who has provided unflagging support to him in all his endeavors for the past fifty-eight years.

Robert L. Norton
Mattapoisett, Mass.
August, 2018

If you find any errors or have comments or suggestions for improvement, please email the author at norton@wpi.edu. Errata as discovered, and other book information, will be posted on the author's web site at <http://www.designofmachinery.com>.

PREFACE

to the First Edition

When I hear, I forget

When I see, I remember

When I do, I understand

ANCIENT CHINESE PROVERB

This text is intended for the kinematics and dynamics of machinery topics which are often given as a single course, or two-course sequence, in the junior year of most mechanical engineering programs. The usual prerequisites are first courses in statics, dynamics, and calculus. Usually, the first semester, or portion, is devoted to kinematics and the second to dynamics of machinery. These courses are ideal vehicles for introducing the mechanical engineering student to the process of design, since mechanisms tend to be intuitive for the typical mechanical engineering student to visualize and create.

While this text attempts to be thorough and complete on the topics of analysis, it also emphasizes the synthesis and design aspects of the subject to a greater degree than most texts in print on these subjects. Also, it emphasizes the use of computer-aided engineering as an approach to the design and analysis of this class of problems by providing software that can enhance student understanding. While the mathematical level of this text is aimed at second- or third-year university students, it is presented *de novo* and should be understandable to the technical school student as well.

Part I of this text is suitable for a one-semester or one-term course in kinematics. Part II is suitable for a one-semester or one-term course in dynamics of machinery. Alternatively, both topic areas can be covered in one semester with less emphasis on some of the topics covered in the text.

The writing and style of presentation in the text are designed to be clear, informal, and easy to read. Many example problems and solution techniques are presented and spelled out in detail, both verbally and graphically. All the illustrations are done with computer-drawing or drafting programs. Some scanned photographic images are also included. The entire text, including equations and artwork, is printed directly from the author's PDF files by laser typesetting for maximum clarity and quality. Many suggested readings are provided in the bibliography. Short problems and, where appropriate, many longer, unstructured design project assignments are provided at the ends of chapters. These projects provide an opportunity for the students *to do and understand*.

The author's approach to these courses and this text is based on over 40 years' experience in mechanical engineering design, both in industry and as a consultant. He has taught these subjects since 1967, both in evening school to practicing engineers and in day school to younger students. His approach to the course has evolved a great deal in that time, from a traditional approach, emphasizing graphical analysis of many structured problems, through emphasis on algebraic methods as computers became available, through requiring students to write their own computer programs, to the current state described above.

The one constant throughout has been the attempt to convey the art of the design process to the students in order to prepare them to cope with *real* engineering problems in practice. Thus, the author has always promoted design within these courses. Only recently, however, has technology provided a means to more effectively accomplish this goal, in the form of the graphics microcomputer. This text attempts to be an improvement over those currently available by providing up-to-date methods and techniques for analysis and synthesis that take full advantage of the graphics microcomputer, and by emphasizing design as well as analysis. The text also provides a more complete, modern, and thorough treatment of cam design than any existing texts in print on the subject.

The author has written three interactive, student-friendly computer programs for the design and analysis of mechanisms and machines. These programs are designed to enhance the student's understanding of the basic concepts in these courses while simultaneously allowing more comprehensive and realistic problem and project assignments to be done in the limited time available than could ever be done with manual solution techniques, whether graphical or algebraic. Unstructured, realistic design problems which have many valid solutions are assigned. Synthesis and analysis are emphasized equally. The analysis methods presented are up to date, using vector equations and matrix techniques wherever applicable. Manual graphical analysis methods are deemphasized. The graphics output from the computer programs allows the student to see the results of variation of parameters rapidly and accurately and reinforces learning.

These computer programs are distributed with this book, and can be run on any Windows NT/2000/XP/Vista/Windows7/8/10 capable computer. Program LINKAGES analyzes the kinematics and dynamics of fourbar, geared fivebar, sixbar, and fourbar slider linkages. It also will synthesize fourbar linkages for two and three positions. LINKAGES also analyzes the slider-crank linkage as used in the internal combustion engine and provides a complete dynamic analysis of single- and multicylinder engine inline, V, and W configurations, allowing the mechanical dynamic design of engines to be done. DYNACAM allows the design and dynamic analysis of cam-follower systems. MATRIX is a general-purpose linear equation system solver. These are student editions of professional programs that are written by the author and that he provides to companies the world over.

All these programs, except MATRIX, provide dynamic, graphical animation of the designed devices. The reader is strongly urged to make use of these programs in order to investigate the results of variation of parameters in these kinematic devices. The programs are designed to enhance and augment the text rather than be a substitute for it. The converse is also true. Many solutions to the book's examples and to the problem sets are downloadable as files to be opened in these programs. Most of these solutions can be animated on the computer screen for a better demonstration of the concept than is possible on the printed page. The instructor and students are both encouraged to take advantage of

The author's intention is that synthesis topics be introduced first to allow the students to work on some simple design tasks early in the term while still mastering the analysis topics. Though this is not the "traditional" approach to the teaching of this material, the author believes that it is a superior method to that of initial concentration on detailed analysis of mechanisms for which the student has no concept of origin or purpose.

Chapters 1 and 2 are introductory. Those instructors wishing to teach analysis before synthesis can leave Chapters 3 and 5 on linkage synthesis for later consumption. Chapters 4, 6, and 7 on position, velocity, and acceleration analysis are sequential and build upon each other. In fact, some of the problem sets are common among these three chapters so that students can use their position solutions to find velocities and then later use both to find the accelerations in the same linkages. Chapter 8 on cams is more extensive and complete than that of other kinematics texts and takes a design approach. Chapter 9 on gear trains is introductory. The dynamic force treatment in Part II uses matrix methods for the solution of the system simultaneous equations. Graphical force analysis is not emphasized. Chapter 10 presents an introduction to dynamic systems modeling. Chapter 11 deals with force analysis of linkages. Balancing of rotating machinery and linkages is covered in Chapter 12. Chapters 13 and 14 use the internal combustion engine as an example to pull together many dynamic concepts in a design context. Chapter 15 presents an introduction to dynamic systems modeling and uses the cam-follower system as the example. Chapter 16 describes servo- and cam-driven linkages. Chapters 3, 8, 11, 13, and 14 provide open-ended project problems as well as structured problem sets. The assignment and execution of unstructured project problems can greatly enhance the student's understanding of the concepts as described by the proverb in the epigraph to this preface.

ACKNOWLEDGMENTS The sources of photographs and other nonoriginal art used in the text are acknowledged in the captions and opposite the title page, but the author would also like to express his thanks for the cooperation of all those individuals and companies who generously made these items available. The author would also like to thank those who reviewed various sections of the first edition of the text and who made many useful suggestions for improvement. Mr. John Titus of the University of Minnesota reviewed Chapter 5 on analytical synthesis and Mr. Dennis Klipp of Klipp Engineering, Waterville, Maine, reviewed Chapter 8 on cam design. Professor William J. Crochetiere and Mr. Homer Eckhardt of Tufts University, Medford, MA., reviewed Chapter 15. Mr. Eckhardt and Professor Crochetiere of Tufts, and Professor Charles Warren of the University of Alabama taught from and reviewed Part I. Professor Holly K. Ault of Worcester Polytechnic Institute thoroughly reviewed the entire text while teaching from the prepublication, class-test versions of the complete book. Professor Michael Keefe of the University of Delaware provided many helpful comments. Sincere thanks also go to the large number of undergraduate students and graduate teaching assistants who caught many typos and errors in the text and in the programs while using prepublication versions. Since the book's first printing, Profs. D. Cronin, K. Gupta and P. Jensen and Mr. R. Jantz have written to point out errors or make suggestions that I have incorporated and for which I thank them. The author takes full responsibility for any errors that may remain and invites from all readers their criticisms, suggestions for improvement, and identification of errors in the text or programs, so that both can be improved in future versions. Contact norton@wpi.edu.

*Robert L. Norton
Mattapoisett, Mass.
August, 1991*

CONTENTS

Preface to the Sixth Editionix

Preface to the First Editionxi

VIDEO CONTENTS.....xxiv

PART I KINEMATICS OF MECHANISMS 1

Chapter 1 Introduction..... 3

1.0	Purpose	3
1.1	Kinematics and Kinetics.....	3
1.2	Mechanisms and Machines	4
1.3	A Brief History of Kinematics.....	5
1.4	Applications of Kinematics.....	6
1.5	A Design Process	7
	<i>Design, Invention, Creativity</i>	7
	<i>Identification of Need</i>	8
	<i>Background Research</i>	9
	<i>Goal Statement</i>	10
	<i>Performance Specifications</i>	10
	<i>Ideation and Invention</i>	10
	<i>Analysis</i>	12
	<i>Selection</i>	13
	<i>Detailed Design</i>	13
	<i>Prototyping and Testing</i>	13
	<i>Production</i>	14
1.6	Other Approaches to Design	15
	<i>Axiomatic Design</i>	15
1.7	Multiple Solutions	16
1.8	Human Factors Engineering.....	16
1.9	The Engineering Report.....	17
1.10	Units.....	17
1.11	A Design Case Study.....	21
	<i>Educating for Creativity in Engineering</i>	21
1.12	What's to Come.....	25
1.13	Resources With This Text.....	26
	<i>Programs</i>	26
	<i>Videos</i>	26
1.14	References.....	27
1.15	Bibliography	27

Chapter 2 Kinematics Fundamentals.....30

2.0	Introduction	30
2.1	Degrees of Freedom (DOF) or Mobility	30
2.2	Types of Motion.....	31
2.3	Links, Joints, and Kinematic Chains	32

2.4	Drawing Kinematic Diagrams	36
2.5	Determining Degree of Freedom or Mobility	37
	<i>Degree of Freedom in Planar Mechanisms</i>	38
	<i>Degree of Freedom (Mobility) in Spatial Mechanisms</i>	40
2.6	Mechanisms and Structures	40
2.7	Number Synthesis	42
2.8	Paradoxes	46
2.9	Isomers	47
2.10	Linkage Transformation	48
2.11	Intermittent Motion	53
2.12	Inversion	53
2.13	The Grashof Condition	55
	<i>Classification of the Fourbar Linkage</i>	60
2.14	Linkages of More Than Four Bars	62
	<i>Geared Fivebar Linkages</i>	62
	<i>Sixbar Linkages</i>	63
	<i>Grashof-Type Rotatability Criteria for Higher-Order Linkages</i>	63
2.15	Springs as Links	65
2.16	Compliant Mechanisms	65
2.17	Micro Electro-Mechanical Systems (MEMS)	67
2.18	Practical Considerations	69
	<i>Pin Joints Versus Sliders and Half Joints</i>	69
	<i>Cantilever or Straddle Mount?</i>	71
	<i>Short Links</i>	71
	<i>Bearing Ratio</i>	71
	<i>Commercial Slides</i>	73
	<i>Linkages Versus Cams</i>	73
2.19	Motors and Drivers	74
	<i>Electric Motors</i>	74
	<i>Air and Hydraulic Motors</i>	79
	<i>Air and Hydraulic Cylinders</i>	79
	<i>Solenoids</i>	79
2.20	References	80
2.21	Problems	81
Chapter 3 Graphical Linkage Synthesis		98
3.0	Introduction	98
3.1	Synthesis	98
3.2	Function, Path, and Motion Generation	100
3.3	Limiting Conditions	102
3.4	Position Synthesis	104
	<i>Two-Position Synthesis</i>	105
	<i>Three-Position Synthesis with Specified Moving Pivots</i>	111
	<i>Three-Position Synthesis with Alternate Moving Pivots</i>	112
	<i>Three-Position Synthesis with Specified Fixed Pivots</i>	114
	<i>Position Synthesis for More Than Three Positions</i>	119
3.5	Quick-Return Mechanisms	119
	<i>Fourbar Quick-Return</i>	119
	<i>Sixbar Quick-Return</i>	121
3.6	Coupler Curves	124
	<i>Symmetrical-Linkage Coupler Curves</i>	129
3.7	Cognates	134
	<i>Roberts-Chebyshev Theorem</i>	134
	<i>Parallel Motion</i>	
	<i>Geared Fivebar Cognates of the Fourbar</i>	

3.8	Straight-Line Mechanisms	141
	<i>Designing Optimum Straight-Line Fourbar Linkages</i>	144
3.9	Dwell Mechanisms	148
	<i>Single-Dwell Linkages</i>	148
	<i>Double-Dwell Linkages</i>	151
3.10	Other Useful Linkages.....	153
	<i>Constant Velocity Piston Motion</i>	153
	<i>Large Angular Excursion Rocker Motion</i>	154
	<i>Remote Center Circular Motion</i>	155
3.11	References.....	157
3.12	Bibliography	158
3.13	Problems	159
3.14	Projects.....	173

Chapter 4 Position Analysis..... 178

4.0	Introduction	178
4.1	Coordinate systems	180
4.2	Position and Displacement	180
	<i>Position</i>	180
	<i>Coordinate Transformation</i>	181
	<i>Displacement</i>	181
4.3	Translation, Rotation, and Complex Motion.....	183
	<i>Translation</i>	183
	<i>Rotation</i>	183
	<i>Complex Motion</i>	184
	<i>Theorems</i>	185
4.4	Graphical Position Analysis of Linkages	185
4.5	Algebraic Position Analysis of Linkages	187
	<i>Vector Loop Representation of Linkages</i>	188
	<i>Complex Numbers as Vectors</i>	188
	<i>The Vector Loop Equation for a Fourbar Linkage</i>	189
4.6	The Fourbar Crank-Slider Position Solution.....	196
4.7	The Fourbar Slider-Crank Position Solution.....	198
4.8	An Inverted Crank-Slider Position Solution	201
4.9	Linkages of More Than Four Bars.....	203
	<i>The Geared Fivebar Linkage</i>	204
	<i>Sixbar Linkages</i>	207
4.10	Position of Any Point on a Linkage	207
4.11	Transmission Angles	208
	<i>Extreme Values of the Transmission Angle</i>	209
4.12	Toggle Positions.....	211
4.13	Circuits and Branches in Linkages.....	212
4.14	Newton-Raphson Solution Method	214
	<i>One-Dimensional Root-Finding (Newton's Method)</i>	214
	<i>Multidimensional Root-Finding (Newton-Raphson Method)</i>	216
	<i>Newton-Raphson Solution for the Fourbar Linkage</i>	217
	<i>Equation Solvers</i>	217
4.15	References.....	218
4.16	Problems.....	218

Chapter 5 Analytical Linkage Synthesis.....233

5.0	Introduction	233
5.1	Types of Kinematic Synthesis.....	233

5.2	Two-Position Synthesis for Rocker Output	234
5.3	Precision Points	236
5.4	Two-Position Motion Generation by Analytical Synthesis	236
5.5	Comparison of Analytical and Graphical Two-Position Synthesis	243
5.6	Simultaneous Equation Solution	245
5.7	Three-Position Motion Generation by Analytical Synthesis	247
5.8	Comparison of Analytical and Graphical Three-Position Synthesis	252
5.9	Synthesis for a Specified Fixed Pivot Location	257
5.10	Center-Point and Circle-Point Circles	263
5.11	Four- and Five-Position Analytical Synthesis	265
5.12	Analytical Synthesis of a Path Generator with Prescribed Timing	266
5.13	Analytical Synthesis of a Fourbar Function Generator	267
5.14	Other Linkage Synthesis Methods	269
	<i>Precision Point Methods</i>	272
	<i>Coupler Curve Equation Methods</i>	273
	<i>Optimization Methods</i>	273
5.15	References	277
5.16	Problems	279

Chapter 6 Velocity Analysis..... 291

6.0	Introduction	291
6.1	Definition of Velocity	291
6.2	Graphical Velocity Analysis	294
6.3	Instant Centers of Velocity	298
6.4	Velocity Analysis with Instant Centers	305
	<i>Angular Velocity Ratio</i>	307
	<i>Mechanical Advantage</i>	309
	<i>Using Instant Centers in Linkage Design</i>	311
6.5	Centroids	313
	<i>A "Linkless" Linkage</i>	316
	<i>Cusps</i>	316
6.6	Velocity of Slip	317
6.7	Analytical Solutions for Velocity Analysis	321
	<i>The Fourbar Pin-Jointed Linkage</i>	321
	<i>The Fourbar Crank-Slider</i>	324
	<i>The Fourbar Slider-Crank</i>	327
	<i>The Fourbar Inverted Crank-Slider</i>	328
6.8	Velocity Analysis of the Geared Fivebar Linkage	330
6.9	Velocity of Any Point on a Linkage	331
6.10	References	333
6.11	Problems	333

Chapter 7 Acceleration Analysis..... 357

7.0	Introduction	357
7.1	Definition of Acceleration	357
7.2	Graphical Acceleration Analysis	360
7.3	Analytical Solutions for Acceleration Analysis	365
	<i>The Fourbar Pin-Jointed Linkage</i>	365
	<i>The Fourbar Crank-Slider</i>	369
	<i>The Fourbar Slider-Crank</i>	371
	<i>Coriolis Acceleration</i>	373
	<i>The Fourbar Inverted Crank-Slider</i>	376

7.4	Acceleration Analysis of the Geared Fivebar Linkage.....	379
7.5	Acceleration of Any Point on a Linkage.....	380
7.6	Human Tolerance of Acceleration	382
7.7	Jerk	385
7.8	Linkages of N Bars	387
7.9	Reference.....	387
7.10	Problems	387
7.11	Virtual Laboratory.....	408

Chapter 8 Cam Design 409

8.0	Introduction	409
8.1	Cam Terminology.....	410
	<i>Type of Follower Motion</i>	<i>410</i>
	<i>Type of Joint Closure.....</i>	<i>412</i>
	<i>Type of Follower.....</i>	<i>413</i>
	<i>Type of Cam</i>	<i>413</i>
	<i>Type of Motion Constraints.....</i>	<i>414</i>
	<i>Type of Motion Program</i>	<i>415</i>
8.2	S V A J Diagrams.....	416
8.3	Double-Dwell Cam Design—Choosing S V A J Functions.....	417
	<i>The Fundamental Law of Cam Design.....</i>	<i>420</i>
	<i>Simple Harmonic Motion (SHM)</i>	<i>421</i>
	<i>Cycloidal Displacement.....</i>	<i>422</i>
	<i>Combined Functions.....</i>	<i>425</i>
	<i>The SCCA Family of Double-Dwell Functions.....</i>	<i>429</i>
	<i>Polynomial Functions.....</i>	<i>439</i>
	<i>Double-Dwell Applications of Polynomials</i>	<i>439</i>
8.4	Single-Dwell Cam Design—Choosing S V A J Functions.....	443
	<i>Single-Dwell Applications of Polynomials</i>	<i>445</i>
	<i>Effect of Asymmetry on the Rise-Fall Polynomial Solution.....</i>	<i>448</i>
8.5	Critical Path Motion (CPM).....	452
	<i>Polynomials Used for Critical Path Motion.....</i>	<i>453</i>
8.6	Sizing the Cam—Pressure Angle and Radius of Curvature	460
	<i>Pressure Angle—Translating Roller Followers</i>	<i>461</i>
	<i>Choosing a Prime Circle Radius.....</i>	<i>464</i>
	<i>Overturning Moment—Translating Flat-Faced Follower</i>	<i>465</i>
	<i>Radius of Curvature—Translating Roller Follower</i>	<i>466</i>
	<i>Radius of Curvature—Translating Flat-Faced Follower.....</i>	<i>470</i>
8.7	Practical Design Considerations	475
	<i>Translating or Oscillating Follower?.....</i>	<i>475</i>
	<i>Force- or Form-Closed?.....</i>	<i>476</i>
	<i>Radial or Axial Cam?.....</i>	<i>476</i>
	<i>Roller or Flat-Faced Follower?.....</i>	<i>476</i>
	<i>To Dwell or Not to Dwell?.....</i>	<i>477</i>
	<i>To Grind or Not to Grind?.....</i>	<i>477</i>
	<i>To Lubricate or Not to Lubricate?.....</i>	<i>478</i>
8.8	References.....	478
8.9	Problems	479
8.10	Virtual Laboratory	485
8.11	Projects.....	485

Chapter 9	Gear Trains	490
9.0	Introduction	490
9.1	Rolling Cylinders	491
9.2	The Fundamental Law of Gearing	492
	<i>The Involute Tooth Form</i>	493
	<i>Pressure Angle</i>	495
	<i>Changing Center Distance</i>	495
	<i>Backlash</i>	497
9.3	Gear Tooth Nomenclature	497
9.4	Interference and Undercutting	500
	<i>Unequal-Addendum Tooth Forms</i>	500
9.5	Contact Ratio	502
9.6	Gear Types	505
	<i>Spur, Helical, and Herringbone Gears</i>	505
	<i>Worms and Worm Gears</i>	506
	<i>Rack and Pinion</i>	507
	<i>Bevel and Hypoid Gears</i>	507
	<i>Noncircular Gears</i>	508
	<i>Belt and Chain Drives</i>	509
9.7	Simple Gear Trains	511
9.8	Compound Gear Trains	512
	<i>Design of Compound Trains</i>	512
	<i>Design of Reverted Compound Trains</i>	514
	<i>An Algorithm for the Design of Compound Gear Trains</i>	517
9.9	Epicyclic or Planetary Gear Trains	520
	<i>The Tabular Method</i>	523
	<i>The Formula Method</i>	527
9.10	Efficiency of Gear Trains	529
9.11	Transmissions	532
9.12	Differentials	536
9.13	References	538
9.14	Bibliography	539
9.15	Problems	539
PART II	DYNAMICS OF MACHINERY	551
Chapter 10	Dynamics Fundamentals	553
10.0	Introduction	553
10.1	Newton's Laws of Motion	553
10.2	Dynamic Models	554
10.3	Mass	554
10.4	Mass Moment and Center of Gravity	556
10.5	Mass Moment of Inertia (Second Moment of Mass)	558
10.6	Parallel Axis Theorem (Transfer Theorem)	559
10.7	Determining Mass Moment of Inertia	560
	<i>Analytical Methods</i>	560
	<i>Experimental Methods</i>	560
10.8	Radius of Gyration	561
10.9	Modeling Rotating Links	562
10.10	Center of Percussion	563

10.11	Lumped Parameter Dynamic Models.....	565
	<i>Spring Constant</i>	566
	<i>Damping</i>	566
10.12	Equivalent Systems	568
	<i>Combining Dampers</i>	569
	<i>Combining Springs</i>	570
	<i>Combining Masses</i>	571
	<i>Lever and Gear Ratios</i>	571
10.13	Solution Methods.....	577
10.14	The Principle of d'Alembert	578
	<i>Centrifugal Force</i>	578
10.15	Energy Methods—Virtual Work	580
10.16	References.....	582
10.17	Problems	582

Chapter 11 Dynamic Force Analysis 589

11.0	Introduction	589
11.1	Newtonian Solution Method.....	589
11.2	Single Link in Pure Rotation.....	590
11.3	Force Analysis of A Threebar Crank-Slide Linkage	593
11.4	Force Analysis of a Fourbar Linkage	599
11.5	Force Analysis of a Fourbar Crank-Slider Linkage.....	606
11.6	Force Analysis of the Inverted Crank-Slider	608
11.7	Force Analysis—Linkages with More Than Four Bars.....	611
11.8	Shaking Force and Shaking Moment.....	612
11.9	Program Linkages.....	613
11.10	Torque Analysis by an Energy Method.....	613
11.11	Controlling Input Torque—Flywheels	616
11.12	A Linkage Force Transmission Index	622
11.13	Practical Considerations	624
11.14	Reference.....	625
11.15	Problems	625
11.16	Virtual Laboratory	638
11.17	Projects.....	639

Chapter 12 Balancing 642

12.0	Introduction	642
12.1	Static Balance	643
12.2	Dynamic Balance	646
12.3	Balancing Linkages	651
	<i>Complete Force Balance of Linkages</i>	651
12.4	Effect of Balancing on Shaking and Pin Forces	655
12.5	Effect of Balancing on Input Torque	656
12.6	Balancing the Shaking Moment in Linkages.....	657
12.7	Measuring and Correcting Imbalance	661
12.8	References.....	663
12.9	Problems	664
12.10	Virtual Laboratory	671

Chapter 13	Engine Dynamics	672
13.0	Introduction	672
13.1	Engine Design	674
13.2	Slider-Crank Kinematics.....	679
13.3	Gas Force and Gas Torque	685
13.4	Equivalent Masses.....	687
13.5	Inertia and Shaking Forces	691
13.6	Inertia and Shaking Torques.....	694
13.7	Total Engine Torque	697
13.8	Flywheels.....	697
13.9	Pin Forces in the Single-Cylinder Engine	699
13.10	Balancing the Single-Cylinder Engine.....	705
	<i>Effect of Crankshaft Balancing on Pin Forces.....</i>	<i>709</i>
13.11	Design Trade-offs and Ratios	709
	<i>Conrod/Crank Ratio.....</i>	<i>710</i>
	<i>Bore/Stroke Ratio.....</i>	<i>710</i>
	<i>Materials.....</i>	<i>710</i>
13.12	Bibliography	711
13.13	Problems	711
13.14	Projects.....	716
Chapter 14	Multicylinder Engines.....	717
14.0	Introduction	717
14.1	Multicylinder Engine Designs.....	719
14.2	The Crank Phase Diagram	722
14.3	Shaking Forces in Inline Engines	723
14.4	Inertia Torque in Inline Engines	727
14.5	Shaking Moment in Inline Engines	728
14.6	Even Firing.....	730
	<i>Two-Stroke Cycle Engine.....</i>	<i>731</i>
	<i>Four-Stroke Cycle Engine.....</i>	<i>733</i>
14.7	Vee Engine Configurations.....	739
14.8	Opposed Engine Configurations.....	751
14.9	Balancing Multicylinder Engines	751
	<i>Secondary Harmonic Balancing of the Four-Cylinder Inline Engine.....</i>	<i>756</i>
	<i>A Perfectly Balanced Two-Cylinder Engine.....</i>	<i>758</i>
14.10	References.....	759
14.11	Bibliography	759
14.12	Problems	760
14.13	Projects.....	761
Chapter 15	Cam Dynamics	763
15.0	Introduction	763
15.1	Dynamic Force Analysis of the Force-Closed Cam-Follower	764
	<i>Undamped Response.....</i>	<i>764</i>
	<i>Damped Response.....</i>	<i>767</i>
15.2	Resonance	772
15.3	Kinetostatic Force Analysis of the Force-Closed Cam-Follower.....	775

15.4	Kinetostatic Force Analysis of the Form-Closed Cam-Follower.....	780
15.5	Kinetostatic Camshaft Torque	783
15.6	Measuring Dynamic Forces and Accelerations	785
15.7	Practical Considerations	789
15.8	References.....	789
15.9	Bibliography	790
15.10	Problems	790
15.11	Virtual Laboratory	794
Chapter 16	Cam- and Servo-Driven Mechanisms	795
16.0	Introduction	795
16.1	Servomotors.....	796
16.2	Servo Motion Control.....	797
	<i>Servo Motion Functions</i>	797
16.3	Cam-Driven Linkages	798
16.4	Servo-Driven Linkages.....	806
16.5	Other Linkages.....	812
16.6	Cam-Driven Versus Servo-Driven Mechanisms	812
	<i>Flexibility</i>	812
	<i>Cost</i>	813
	<i>Reliability</i>	813
	<i>Complexity</i>	813
	<i>Robustness</i>	813
	<i>Packaging</i>	814
	<i>Load Capacity</i>	814
16.7	References.....	815
16.8	Bibliography	815
16.9	Problems	815
Appendix A	Computer Programs.....	817
Appendix B	Material Properties.....	819
Appendix C	Geometric Properties.....	823
Appendix D	Spring Data.....	825
Appendix E	Coupler Curve Atlases	829
Appendix F	Answers to Selected Problems.....	831
Appendix G	Equations for Under- or Overbalanced Multicylinder Engines.....	847
Index.....		851
Downloads Index.....		867

VIDEO CONTENTS

The Sixth Edition has a collection of *Master Lecture Videos and Tutorials* made by the author over a thirty-year period while teaching at Worcester Polytechnic Institute. The lectures were recorded in a classroom in front of students in 2011/2012. Tutorials were done in a recording studio and were intended as supplements to class lectures. Some have become lectures here.

There are 82 instructional videos in total. Thirty-four are "50-minute" lectures and are labeled as such. Thirty-nine are short "snippets" from the lecture videos that are linked to the relevant topics in a chapter. Seven are demonstrations of machinery or tutorials. Two are laboratory exercises that have been "virtualized" via video demonstration and the provision of test data so that students can simulate the lab. The run times of all videos are noted in the tables.

The sixth edition is available both as a print book and as digital media. The digital, e-book versions have active links that allow these videos to be run while reading the book. The print edition notes the names and URLs of all the videos in the text at their links.

In addition to the lecture videos, all the digital content that was with the fifth and earlier editions is still available as downloads, including the author-written programs LINKAGES, DYNACAM, and MATRIX. An index of all these files is in the **Downloads Index** and includes 105 video animations of figures in the book. In the digital e-book versions, these are hotlinked to the figures that they animate. The URL of each figure video is also provided in the figure for print-book readers to download them.

Because this book is about the study of motion, it is particularly well suited to digital media. The figure animations have been in the book since the third edition as files on a DVD. It is unknown how many students bothered to open and run those files. The digital versions of the book will make this much easier to do.

Any instructor or student who uses the print book may register on my website, <http://www.designofmachinery.com>, either as a student or instructor, and I will send them a password to access a protected site where they can download the latest versions of my computer programs, LINKAGES, DYNACAM, and MATRIX. They can also view the 82 videos and download all the files listed in the Download Index. Note that I personally review each of these requests for access and will approve only those that are filled out completely and correctly according to the instructions. I require complete information and only accept university email addresses.

LECTURE VIDEOS AND SNIPPETS Part 1

(Concatenate this URL with any filename below to run a video)

Chapter	Lecture	Snippet	Topic	http://www.designofmachinery.com/DOM/	Run Time
1	1		Introduction	Introduction.mp4	39:10
1	2		Design Process and Documentation	Design_Process_and_Documentation.mp4	43:28
1		2A	<i>Design Process</i>	<i>Design_Process.mp4</i>	29:46
1		2B	<i>Documentation</i>	<i>Documentation.mp4</i>	15:57
1		2C	<i>Units of Mass</i>	<i>Units.mp4</i>	10:06
2	3		Kinematics Fundamentals	Kinematics_Fundamentals.mp4	49:12
2		3A	<i>Degree of Freedom</i>	<i>DOF.mp4</i>	03:53
2		3B	<i>Links and Joints</i>	<i>Links_and_Joints.mp4</i>	11:00
2		3C	<i>Grubler</i>	<i>Links_and_Joints.mp4</i>	14:29
2		3D	<i>Number Synthesis</i>	<i>Number_Synthesis.mp4</i>	03:47

LECTURE VIDEOS AND SNIPPETS Part 2

(Concatenate this URL with any filename below to run a video)

Chapter	Lecture Snippet	Topic	http://www.designofmachinery.com/DOM/	Run Time
2	3E	Isomers	<i>Isomers.mp4</i>	04:15
2	3F	Inversion	<i>Inversion.mp4</i>	03:40
2	3G	Grashof Condition	<i>The_Grashof_Condition.mp4</i>	07:21
2	3H	Fivebar Linkage	<i>Fivebar.mp4</i>	01:33
2	3I	Compliant Linkages	<i>Compliant_Linkages.mp4</i>	01:27
3	4	Position Synthesis	Position_Synthesis.mp4	47:57
3	5	Quick Return Linkages	Quick_Return_Linkages.mp4	55:22
3	6	Coupler Curves and Atlases	Coupler_Curves.mp4	59:57
3	7	Symmetrical and Straight Coupler Curves	Symmetrical_and_Straight_Coupler_Curves.mp4	14:47
3	7A	<i>Symmetrical Coupler Curves</i>	<i>Symmetrical_Coupler_Curves.mp4</i>	05:47
3	7B	<i>Straight Line Linkages</i>	<i>Straight_Line_Linkages.mp4</i>	09:20
3	8	Cognates of Linkages	Cognates_of_Linkages.mp4	18:12
3	9	Parallel Motion	Parallel_Motion.mp4	21:49
3	10	Dwell Linkages	Dwell_Mechanisms.mp4	35:36
4	11	Position Analysis	Position_Analysis.mp4	49:48
5	12	Analytical Linkage Synthesis	Analytical_Linkage_Synthesis.mp4	48:17
6	13	Instant Centers and Centroids	Instant_Centers_and_Centroids.mp4	49:16
6	13A	<i>Instant Centers</i>	<i>Instant_Centers_Tutorial.mp4</i>	28:55
6	13B	<i>Centroids</i>	<i>Centroids.mp4</i>	21:01
6	14	Velocity Analysis with ICs	Velocity_Analysis_with_ICs.mp4	49:48
6	15	Velocity Analysis with Vectors	Velocity_Analysis_with_Vectors.mp4	46:41
7	16	Acceleration Analysis	Acceleration_Analysis.mp4	41:39
7	Lab	Fourbar Virtual Laboratory Video	Fourbar_Machine_Virtual_Laboratory.mp4	35:38
7	Lab	Fourbar Virtual Laboratory Data	Fourbar_Virtual_Lab.zip	–
8	17	Cam Design I	Cam_Design_I.mp4	50:42
8	18	Cam Design II	Cam_Design_II.mp4	51:16
8	19	Cam Design III	Cam_Design_III.mp4	48:54
8	Lab	Cam Machine Virtual Laboratory Video	Cam_Machine_Virtual_Laboratory.mp4	21:28
8	Lab	Cam Machine Laboratory Data	Cam_Virtual_Lab.zip	–
9	20	Gear Design	Gear_Design.mp4	54:45
9	21	Gear Trains	Gear_Trains.mp4	37:53
9	22	Gear Transmissions	Gear_Transmissions.mp4	41:06
10	23	Dynamics Fundamentals	Dynamics_Fundamentals.mp4	52:01
10	23A	<i>Newtons Laws</i>	<i>Newtons_Laws.mp4</i>	04:00
10	23B	<i>Units of Mass</i>	<i>Mass.mp4</i>	10:06
10	23C	<i>Moments of Mass</i>	<i>Moments_of_Mass.mp4</i>	04:33
10	23D	<i>Transfer Theorem</i>	<i>Transfer_Theorem.mp4</i>	02:15
10	23E	<i>Radius of Gyration</i>	<i>Radius_of_Gyration.mp4</i>	01:21
10	23F	<i>Rotating Links</i>	<i>Rotating_Links.mp4</i>	02:11
10	23G	<i>Center of Percussion</i>	<i>Center_of_Percussion.mp4</i>	06:58
10	23H	<i>Equivalent Systems</i>	<i>Equivalent_Systems.mp4</i>	14:53
10	23I	<i>Lumped Models</i>	<i>Lumped_Models.mp4</i>	19:39
10	23J	<i>Lever Ratios</i>	<i>Lever_Ratios.mp4</i>	05:16
10	23K	<i>Modeling Systems</i>	<i>Modeling_Systems.mp4</i>	06:44
10	23L	<i>D'Alembert Force</i>	<i>D'Alembert_Force.mp4</i>	03:36
10	23M	<i>Centrifugal Force</i>	<i>Centrifugal_Force.mp4</i>	06:58

LECTURE VIDEOS AND SNIPPETS Part 3

(Concatenate this URL with any filename below to run a video)

Chapter	Lecture	Snippet	Topic	http://www.designofmachinery.com/DOM/	Run Time
11	24		Dynamic Force Analysis	Dynamic_Force_Analysis.mp4	27:28
11		24A	<i>Single Link in Rotation</i>	<i>Single_Link_in_Rotation.mp4</i>	15:29
11		24B	<i>Fourbar Force Analysis</i>	<i>Fourbar_Force_Analysis.mp4</i>	12:18
11	Lab		Fourbar Virtual Laboratory Video	Fourbar_Machine_Virtual_Laboratory.mp4	35:38
11	Lab		Fourbar Virtual Laboratory Data	Fourbar_Virtual_Lab.zip	–
11	25		Virtual Work and Flywheels	Virtual_Work_and_Flywheels.mp4	34:50
11		25A	<i>Virtual Work</i>	<i>Virtual_Work.mp4</i>	10:52
11		25B	<i>Flywheels</i>	<i>Flywheels.mp4</i>	24:07
12	26		Balancing	Balancing.mp4	48:09
12		26A	<i>Static Balance</i>	<i>Static_Balance.mp4</i>	09:58
12		26B	<i>Dynamic Balance</i>	<i>Dynamic_Balance.mp4</i>	09:42
12		26C	<i>Linkage Balancing</i>	<i>Linkage_Balancing.mp4</i>	26:55
12		26D	<i>Field Balancing</i>	<i>Field_Balancing.mp4</i>	02:43
12	Lab		Fourbar Virtual Laboratory Video	Fourbar_Machine_Virtual_Laboratory.mp4	35:38
12	Lab		Fourbar Virtual Laboratory Data	Fourbar_Virtual_Lab.zip	–
13	27		Engine Kinematics	Engine_Kinematics.mp4	48:17
13	28		Engine Dynamics	Engine_Dynamics.mp4	53:17
13	29		Engine Balancing and Pin Forces	Engine_Balancing_and_Pin_Forces.mp4	42:38
13		29A	<i>Pin Forces</i>	<i>Pin_Forces.mp4</i>	20:03
13		29B	<i>Dynamic Balance</i>	<i>Balancing_One_Cylinder.mp4</i>	21:46
14	30		Multicylinder Engines	Multicylinder_Engines.mp4	44:25
14	31		Even Firing	Even_Firing.mp4	47:29
14	32		Vee Engines	Vee_Engines.mp4	48:25
14	33		Balancing Multicylinders	Balancing_Multicylinders.mp4	31:12
15	34		Cam Dynamics	Cam_Dynamics.mp4	48:29

DEMONSTRATION VIDEOS

(Concatenate this URL with any filename below to run a video)

Topic	http://www.designofmachinery.com/DOM/	Run Time
Boot Testing Machine	Boot_Tester.mp4	19:02
Bottle Printing Machine	Bottle_Printing_Machine.mp4	09:38
Grashof Condition	The_Grashof_Condition.mp4	24:12
High-Speed Spring Failure	Spring_Failure.mp4	03:46
Pick and Place Mechanism	Pick_and_Place_Mechanism.mp4	38:35
Spring Manufacturing Machinery	Spring_Manufacturing.mp4	12:23
Vibration Testing	Vibration_Testing.mp4	05:51

Note that you can download a PDF file containing hyperlinks to all the video content listed in the above tables. This allows print-book readers to easily access the videos without having to type in each URL as noted in the tables. Download the file:

http://www.designofmachinery.com/DOM/Video_Links_for_DOM_6ed.pdf